Interactive School Navigation Dashboard — Rawalpindi

School location selection, geolocation and road-route planning

STUDENT NAME	ARID NUMBER
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ASSIGNED CITY	SELECTED SCHOOL
Rawalpindi, Punjab, Pakistan	Saint John's High School

1. Objective

To develop an interactive WebGIS dashboard for viewing selected schools in Rawalpindi and finding a route from the user's current location to a selected school. The dashboard uses HTML, CSS, JavaScript, Leaflet.js, OpenStreetMap, browser geolocation, and a routing service.

2. Selected Locations

School	Address
Saint John's High School	127 Murree Road, Saddar, Rawalpindi
Government Denny's Higher Secondary School	Kashmir Road, Saddar, Rawalpindi

The dashboard includes the supplied school addresses and uses map markers to represent the locations. Confirm the markers against the address before submitting if the public geocoding or map service returns a different point.

3. Dashboard Features

- OpenStreetMap base map displayed with Leaflet.js.
- Two school choices in the left sidebar.
- Student name and ARID number fields in the right sidebar.
- Browser geolocation to obtain the user's current position, with permission.
- Route line, distance in kilometres, and estimated driving time through the OSRM demo routing service.

4. How to Run the Dashboard

• Save the accompanying HTML file to your computer.

• Open it in a modern browser while connected to the internet. If geolocation is blocked, run it from VS Code using a local development server such as Live Server.

• The student name and ARID number are already entered in the right panel.

• Select a school, click **Use My Current Location**, allow browser location permission, then click **Find Route to Selected School**.

• Take a screenshot of the live dashboard and a second screenshot after a route has been calculated.

5. Conceptual Questions and Answers

Q1. What is the difference between OpenStreetMap and Leaflet.js in this WebGIS application?

OpenStreetMap provides the map data and map tiles. Leaflet.js is the JavaScript library used to display and interact with the map, markers, popups, and route line in the browser.

Q2. How does the browser obtain the user's current location?

The application calls the browser Geolocation API. The browser asks the user for permission and, if permission is granted, provides latitude and longitude.

Q3. What happens when the user clicks a location from the sidebar?

The selected school becomes active, the map moves to its marker, and the school popup is shown. The user can then request a route from their current location.

Q4. Why is a Routing API required in this dashboard?

A routing service calculates a route along the road network and returns route geometry, distance, and estimated travel time. A straight line between two points would not provide road directions.

Q5. What information is required to locate a place on a WebGIS map?

A place can be located using geographic coordinates (latitude and longitude). A place name or address can also be submitted to a geocoding service to obtain coordinates.

6. Important Code

The HTML dashboard contains the main implementation. These are the key components:

```
L.map('map').setView([33.598, 73.054], 14);
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
   attribution: '&copy; OpenStreetMap contributors'
}).addTo(map);

navigator.geolocation.getCurrentPosition(successCallback, errorCallback);
L.Routing.control({ waypoints: [startPoint, destination] }).addTo(map);
```

7. Dashboard Screenshot

Dashboard preview included in this report. For final submission, also capture the dashboard running in your browser and a genuine route result after the online map and routing services load.

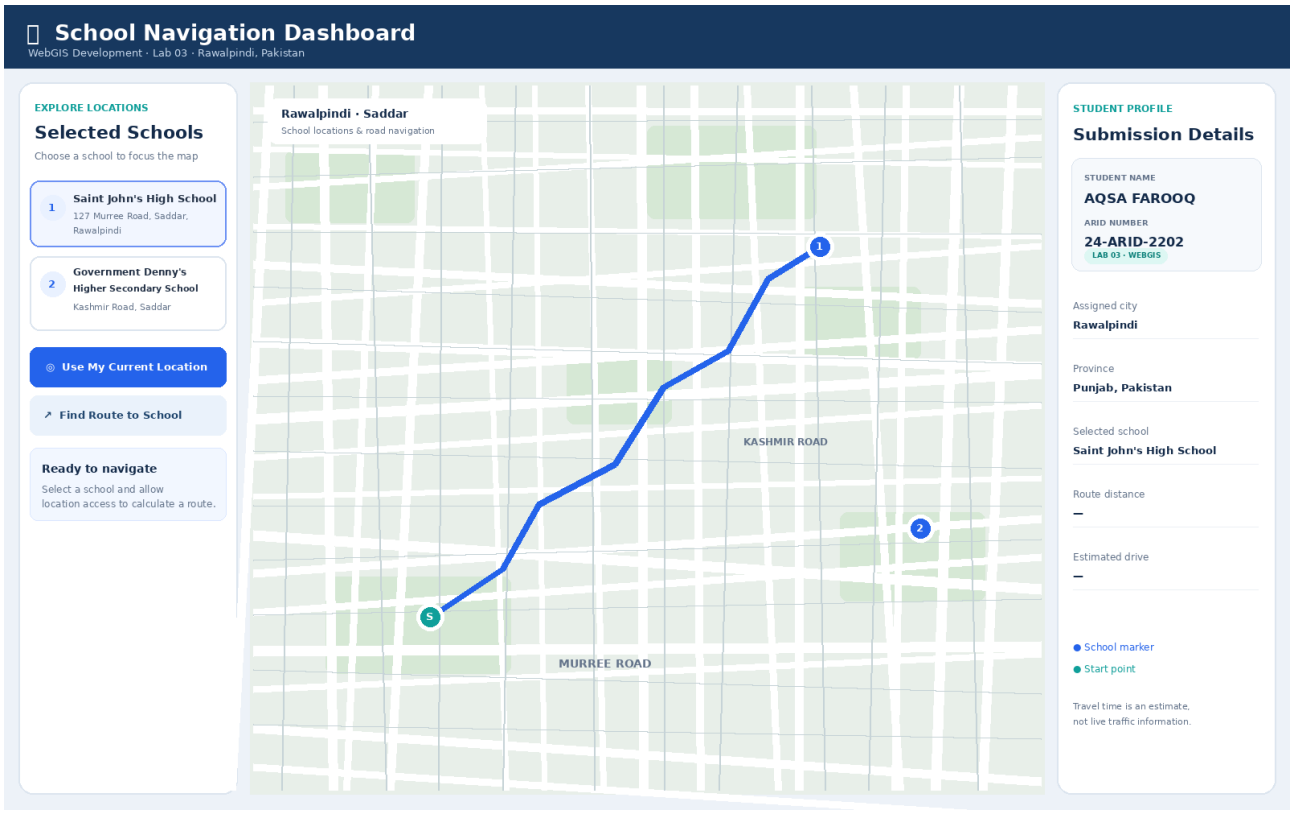


Figure 1. Interactive School Navigation Dashboard with student details and selected school markers.

8. AI Use and Required Screenshots

AI assistance was used to draft the HTML, CSS, and JavaScript structure, add browser geolocation, connect a routing service, and prepare conceptual explanations. The assignment requires at least two screenshots of AI interaction. Capture genuine screenshots of your AI conversation and insert them into the final PDF. Also insert a dashboard screenshot and a route screenshot after running the HTML file.

Submission checklist

- ■ Student name and ARID number
- ■ Assigned city
- ■ Selected schools
- ■ Dashboard screenshot
- ■ Route screenshot
- ■ Important code
- ■ At least two genuine AI-interaction screenshots
- ■ Answers to all five conceptual questions

Note: Internet access is required for map tiles, geocoding, and routing. The OSRM demo service and public geocoding service are for demonstration and may be rate-limited or unavailable. Route time is an estimate, not live traffic information.

Screenshot note: the dashboard image in this report is a visual preview of the submitted HTML layout. The live route screenshot and genuine AI-interaction screenshots must be captured from the actual browser/conversation to meet the original submission checklist.